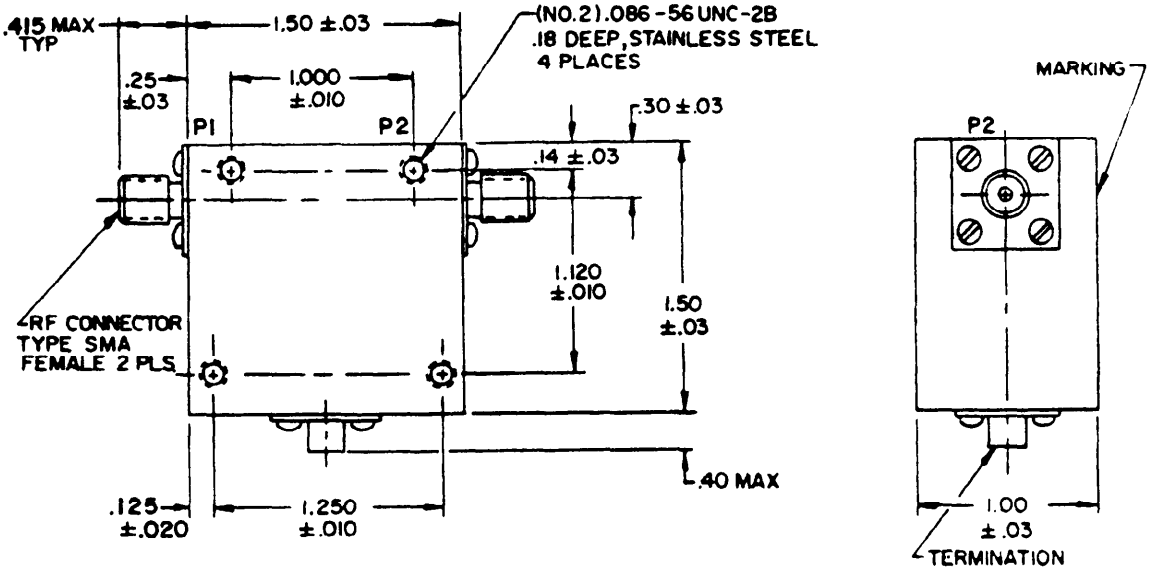
MILITARY SPECIFICATION SHEET

ISOLATOR, RADIO FREQUENCY,
COAXIAL FREQUENCY RANGE - .85 TO 1.05 GHz

This specification sheet is approved for use by U.S. Army Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the isolator described herein shall consist of this document and the latest issue of Specification MIL-I-28791.



INCHES	MM
1.00	25.40
1.50	38.10

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

FIGURE 1. Dimensions and configuration.

FSC 5985

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:49Z
Check the source to verify that this is the current version before use.

ENGINEERING DATA:

Frequency range: .85 to 1.05 GHz.

Nominal impedance: 50 ohms.

Ambient temperature:
Operating - -45°C to +71°C.
Storage and nonoperating - -65°C to +71°C.

Peak power rating: 1,000 watts input with a duty cycle of .002, 2 watts termination.

VSWR maximum load: 2.5:1.

Input port: P1 shall be stainless steel and shall mate with SMA male in accordance with MIL-C-39012.

Output port: P2 shall be stainless steel and shall mate with SMA male in accordance with MIL-C-39012.

REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 4.5 ounces maximum.

Dielectric withstanding voltage: 1,000 volts rms minimum.

VSWR: 1.25:1 maximum.

Isolation: 20 dB minimum.

Insertion loss:
At 25°C - 0.4 dB maximum.
Over temperature range - 0.5 dB maximum.

Shock (specified pulse): In accordance with MIL-I-28791, except test condition G.

Vibration, high frequency: Not applicable. Isolators shall be tested in accordance with method 201 of MIL-STD-202.

Marking: See figure 1.
Part number: M28791/4-01.

NOTE: The isolator covered by this specification sheet is substitutable for the following commercial types:

Part number	Manufacturer's code	Commercial type
M28791/4-01	53720	AER-0141
M28791/4-01	80063	SM-C-801267

Custodian:
Army - EL

Review activities:
Army -
DSA - ES

User activity:
Army -MU

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5985-A452)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:49Z
Check the source to verify that this is the current version before use.

To detach this f cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION MIL-I-28791/4(EL) ISOLATOR, RADIO FREQUENCY, COAXIAL FREQUENCY RANGE - .85 TO 1.05 GHz

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 86 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:49Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army
Headquarters
U.S.Army Electronics Command
Fort Monmouth, New Jersey 07703

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DOD-316

U.S. MAIL

OFFICIAL BUSINESS

Commanding General
U.S Army Electronics Command
ATTN: DRSEL-RD-TS
Fort Monmouth, New Jersey 07703

To detach this form, cut along this line

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:49Z
Check the source to verify that this is the current version before use.